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Improving pension information: Experimental evidence on learning using online resources

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ABSTRACT

Deciding what to do with one's pension funds is a high-stakes, one-shot decision. Retirement schemes are often described in technical jargon that few people understand. We consider whether the learning process can be eased by providing information in video format (vs. the standard textual format) and by changes to the user interface of the websites on which individuals learn about their pension options. The results of a field experiment with 50 to 70-year-old participants and a laboratory experiment with university students both indicate that videos are significantly and substantively more effective in increasing the number of correct responses to a retirement comprehension test. The effect of changing the format from text to video on the number of correct responses is comparable to the effect of a large improvement in financial literacy. The significance of this effect is robust to changes in the content of the videos, with equivalent impact in private and public pension schemes. The study is conducted in association with the Chilean pension authority (*Superintendencia de Pensiones*) and experts from the National Social Security Organization of Chile (*Instituto de Previsión Social*).

1. Introduction

Choosing a retirement scheme is a one-shot, high-stakes decision. In many countries, individuals must decide what to do with their private pension funds and ensure that they receive the public benefits they are eligible for. The Chilean pension system is no exception.³ The information available is often presented in technical language that few people understand and in a format that many find daunting. It is, therefore, not surprising that many people navigating the private pension scheme resort to paying for advice (Granados et al., 2019) or leave money on the table by not applying for public benefits (Centro UC et al., 2017).

Governments are responsible for providing effective communication (OECD, 2017; UNDP, 2018). In the context of pensions, this means presenting information in a way that increases comprehension of the decisions needed to retire. One action public organizations can take is to improve how information is presented on their websites. Many pension-related websites display information entirely in textual format and are structured in terms of products with technical names such as Annuities, Programmed Withdrawals, Beneficiaries, and Online Applications.

It is not clear which changes to the choice architecture (user interface) of a website, if any, are effective at helping people understand the differences between alternative retirement schemes and the public

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³ For more details about the Chilean Pension System, see Appendix Section A.1.

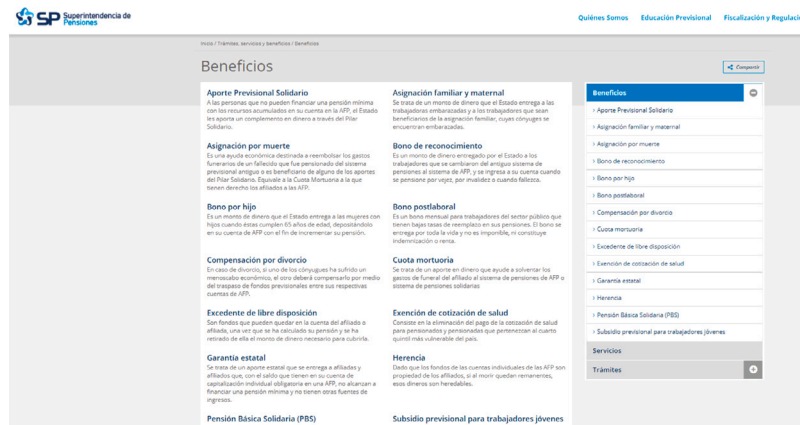


Fig. 1. Screen shot of the official Chilean pensions superintendency retirement advice website in 2019.

benefits available — See Fig. 1 for an example of the status quo. The education literature suggests that innovative ways of presenting information, including videos, can be powerful tools for teaching school-age children (see Karppinen, 2005; Berk, 2009, among others), as well as the general population (Mayer and Moreno, 2003). The choice architecture literature suggests that reducing the number of elements in a choice task (home screen) makes engaging with the information on the website more likely, improves the user experience, and fosters learning (Johnson et al., 2012; Hoffmann et al., 2019; Stone and University, 2005; Um et al., 2012; Heidig et al., 2015). However, little of this has been evaluated on a population of decision-makers over 50 years of age who may not be as comfortable with audiovisual interactions online as younger generations (Prescott, 2021; Anderson and Perrin, 2017).

In this study, we conduct an experimental evaluation of two innovations to the status quo using a 2*2 design: (1) changing the medium of description by presenting the most relevant information in video format instead of text, while keeping the wording constant; and (2) changing the structure of the homepage (the choice task) from long to short by restructuring the information in terms of pensioner profile types instead of the current long list of pension products (see Johnson et al., 2012). To ensure the robustness of the content of the websites, we implemented two variations: (1) one providing information on how to retire using funds in private savings accounts, and (2) one with information on how to access available public benefits. The study thus consists of a total of eight experimental variations, encompassing two format changes, two structural changes, and two content differences. The format and structural changes are the treatment factors in our experimental design, while the two different content variations are best thought of as different arenas in which the 2×2 treatment design is applied.

We conducted both an online field experiment with a sample of 458 participants aged 50 or above who have not yet retired, as well as a fully randomized laboratory version of the experiment with a convenience sample of 209 students. Using two samples helps establish the robustness of any results that we obtain and reduces potential replication or measurement errors (see Levitt and List, 2007; Camerer et al., 2016; Duch et al., 2020). The online sample allows us to reach the population of interest whom a change in the information structure would directly affect. Older decision-makers are generally more difficult to reach, and only a few studies have attempted to do so (Duch et al., 2021; Nam and Loibl, 2021; Xue et al., 2020; de Grip et al., 2020; Pereira and Afonso, 2020).

The experiment was conducted in Chile, and all the content was constructed in collaboration with experts from the Chilean Pension Authority *Superintendencia de Pensiones* and the National Social Security Organization *Instituto de Previsión Social*. Replication material is

available on GitHub.⁴ Our results show that the change from text to video produced a significant increase in comprehension of the relevant aspects of the retirement decision-making process, as measured by a quiz that participants took.⁵ In contrast, the change in format from long-product-oriented to short-profile-structured did not improve understanding. The significance and magnitude of the effects are consistent across public or private pension tasks, as well as across gender and level of financial literacy of participants. The results also are very similar between the laboratory and online implementations. One potential reason why the profile structure did not produce the intended effect is that, when combined with a video, it is associated with a decrease in participants' levels of emotional arousal, which may reflect reduced attention to the information.

While the specific application of our study is the Chilean pension system, the decision situation is quite general. Individuals in OECD and many other countries often face a choice between private and public retirement options, and this decision is one of the most consequential financial decisions of their lives. While the specifics of how the private and pension plans may differ among different national plans, we believe that our conclusions are quite general: that video presentation improves comprehension and that profile information does not. Indeed, the Chilean pension system has been of particular interest to economists and policymakers for the last few decades, since the country enacted the world's first national private pension system in 1981. Many countries have copied the Chilean system, including more than 30 countries across Latin America, Southeast Asia and Eastern Europe.

In our view, this paper contributes to the literature on financial literacy and education. Previous literature on financial literacy has established associations between retirement planning and knowledge of interest compounding, inflation, and risk diversification (Lusardi and Mitchell, 2007; Lusardi and Mitchell, 2008, 2011a). While prior studies have largely focused on the retirement planning stage, we focus on a new decision point — what to do after planning ends and the individual actually retires. We test two interventions to be implemented at this decision point: (1) a short video (following Ambuehl et al., 2014; Lusardi et al., 2017); and (2) a redesigned homepage structure (choice architecture), comparing a single-page layout with all options presented against an organized layout by user profiles.

Our results contribute to the education literature by providing some evidence on what does and does not work to improve comprehension of complex material. While the profile treatment does not improve

⁴ The repository is available at <https://github.com/deniselaroze/Pensions-Website-Design>.

⁵ The videos are available at <https://www.youtube.com/channel/UCmWewgPWNH037JPSUzAqfg>.

retirement literacy, the video did increase comprehension. The results of the videos provide evidence that short informative videos can effectively convey complex information in an age group not previously studied (see [Karppinen, 2005](#); [Berk, 2009](#); [Mayer and Moreno, 2003](#), among many others). The failure of the profile intervention, on the other hand, underscores the fact that choice architecture may not always yield the desired outcomes ([Johnson et al., 2012](#); [Hoffmann et al., 2019](#)). Additionally, our use of Facereading technology allows us to highlight the emotional responses users may have to interface changes ([Stone and University, 2005](#); [Um et al., 2012](#); [Heidig et al., 2015](#)).

The sections that follow provide a review of the literature on the main components of the study, such as financial literacy, decision architecture, and emotions when interacting with web platforms. Subsequently, we present the design and procedure of the study, followed by its main results, and conclude with the discussion and final remarks.

2. Related literature

2.1. Financial literacy

Financial illiteracy is generally widespread across demographically representative populations ([Lusardi and Mitchell, 2014](#)).⁶ In particular, knowledge about pension-related topics is limited, with women, older adults, and individuals with lower levels of education scoring lower than other groups ([Hasler and Lusardi, 2017](#)).⁷ Financially illiterate individuals are also less likely to plan for retirement ([Lusardi and Mitchell, 2011b](#)). [Hoffmann and Otteby \(2018\)](#) suggest that the consumers most likely to engage with informative finance blogs are those who may need them the least, as they tend to have higher financial literacy and face less perceived financial uncertainty.

Chile is no exception to this pattern. [Landerretche and Martínez \(2013\)](#) find that Chileans have limited knowledge regarding the pension system, its rules, and the consequences of their (in)decisions. Their results also suggest that the subgroups with higher levels of financial literacy behave more financially responsibly concerning pension savings. They are more likely to have additional, voluntary savings, display active financial behavior, and voluntarily contribute to their pension funds when self-employed.

Learning about the retirement process has a cost in terms of effort. One has to understand what the retirement options are and how to access them. Middle- and high-income earners can side-step this information hurdle by paying a third-party specialist for advice and reducing the effort to a monetary cost. On the other hand, low-income earners who would like to access public benefits have to do so on their own. Even though there are national information offices that can give guidance regarding what benefits a person is eligible for, an individual has to choose whether or not to make the effort to apply. Failing to apply for public benefits, when one is eligible, is a clear welfare loss to the individual. The difficulty is in getting past the first hurdle of understanding what benefits are available and if one is a potential beneficiary.

⁶ In a measure of financial literacy composed of three questions, one on interest rate compounding, another on inflation, and a last one on stock risk — only 30.2% of American respondents, and a similar proportion in France, answered all three questions correctly. In Germany (53.2%), Switzerland (50.1%), and the Netherlands (44.8%) the proportions were somewhat higher.

⁷ The gender difference seems to be more acute in some countries. For example, [Potrich et al. \(2018\)](#) observe that financial literacy is lower among women than men in their Brazilian sample. On the other hand [Grohmann et al. \(2021\)](#) do not detect any gender differences in financial literacy among middle-class individuals in Bangkok, Thailand.

2.2. Choice architecture

When it comes to decisions as crucial as selecting a retirement scheme, official institutional websites are a key source of information in the retirement decision-making process. They are typically considered a reputable source of information that people can rely on for objective information that is up to date with the most recent policy changes. [Hentzen et al. \(2023\)](#) explored the impact of digital technology in Australia's Superannuation Industry on communication within the sector. The findings revealed that complex retirement information leads to member confusion and disengagement. Consumers increasingly favor self-service approaches, which limits Superfunds' communication efforts. Indeed, sometimes the usual website is so loaded with information that it has all the characteristics of a "Paradox of Choice" ([Iyengar and Lepper, 2001](#); [Schwartz, 2004](#); [Scheibehenne et al., 2010](#); [Kinjo and Ebina, 2015](#)), with the large number of options hindering the making of good choices — [Fig. 1](#) presents a screenshot of the status quo at the time of this study and highlights the density of information currently on display.

Choice architecture can offer some options to address this issue by changing the format in which information is provided — a.k.a. the user experience of the websites ([Stone et al., 2005](#)). One possible alternative is to reduce the effort of acquiring information by providing key concepts in video that complements audio and images.

The education literature suggests that innovative ways of presenting information, including videos, can be a powerful tool for teaching school-age children (see [Karppinen, 2005](#); [Berk, 2009](#), among others) and general populations ([Mayer and Moreno, 2003](#)). Governments, NGOs, and other institutions have already started to use short informative videos as part of their educational material. [Lusardi et al. \(2017\)](#) study the impact of different online financial literacy training tools and find that an informational brochure, a visual interactive tool, a written narrative, and a video narrative are all effective tools in increasing self-efficacy. However, videos, in particular, improved financial literacy more effectively than written narratives or any of the other tools. Specifically, video narratives yielded a significant increase in correct responses (OLS coefficient = 0.0990) and a significant decrease in "I don't know" responses (OLS coefficient = -0.0938), both significant at $p < 0.01$ ($n = 889$). For example, the tutorials of the UK's National Health Service (NHS), the [WorldBank](#), and [Unicef](#) are strongly image- and video-based.⁸

Nevertheless, not every format change impacts learning about pensions. [Hoffmann and Plotkina \(2020\)](#), in a longitudinal study of 736 U.S. participants, examined whether the source (government vs. peer-generated), tone (prescriptive vs. descriptive), and presence of graphical illustrations (vs. text-only) impact consumers' intentions to plan for retirement. The study found that the message format did not affect its effectiveness, but information from a government source was more effective than peer-generated content in increasing consumers' willingness to learn about retirement planning. Specifically, not all videos or e-learning facilities are useful ([Lim Fat et al., 2011](#)). An evaluation of a multi-method e-learning innovation designed to teach clinical skills to student nurses finds that students do appreciate the videos, but also notes that they do not improve performance outcomes on their own ([Kelly et al., 2009](#)).

Thus, despite some positive findings, it is not evident that a visual format is always an efficient way of transmitting information to an older population, especially when there are relevant consequences at

⁸ [Wang-Ly et al. \(2022\)](#) focused on ways to improve retirement pension fund decisions, as we do here. In their experiment, participants engaged in 40 trials where they chose among four fictitious retirement funds. They tested the effects of defaults, disclosure of fees, advice, and the availability of two types of calculators, a Basic and a Smart Calculator, on decisions. They found that advice and the Smart Calculator improved pension plan decisions.

stake (Taylor and Thompson, 1982). This age group is not as used to technology as younger generations, and reading text may be a more natural mode of self-study for them. It may be that despite the availability of visual aids, the increased ease of acquiring information is not sufficient to substantively reduce the use of pension advisors or sales agents. If that is the case, it would not be cost-effective to produce images and video content. It is, therefore, relevant to study the impact of this innovation on an older age group.

A second alternative for easing the cognitive effort required to navigate retirement information is to change the description of the choice task by reducing the number of icons available on the homepage (see Johnson et al., 2012). This can be achieved by structuring the retirement information in terms of user profiles, allowing individuals to select the profile appropriate for them. The profiles aggregate relevant products for certain types of people and exclude those unlikely to apply. This simplifies the homepage choice set, limiting the likelihood of a choice paradox, by reducing the cognitive load needed to engage (Agnew and Szykman, 2005; Townsend and Kahn, 2013), and facilitating learning. Of course, introducing this simplification involves a trade-off, because modifying a website's homepage is costly and requires the collaboration of a multidisciplinary group of experts to select profiles, organize information, and implement changes.

It is essential to recognize that individuals are not passive recipients of a predetermined architecture; they actively engage with and respond to it based on their unique attributes and circumstances (Krijnen et al., 2017). Therefore, choice architects must understand this intricate process and test their interventions before large-scale implementation, as any reduction in the choice set not accepted by users could have unintended consequences.

2.3. Emotions

The literature on user interfaces and user experience suggests that emotions play a strong role in how people interact with different websites. Forlizzi and Battarbee (2004) state that emotions can explain our disposition to act, the valence of that disposition, and the face and body changes associated with arousal. Emotions are also related to the evaluation of the outcomes and perceptions around a process of interaction with a product. Thus, the authors claim that, when the outcome is satisfactory, there is a feeling of accomplishment, and effort is reduced, supporting a more fluent experience. However, if the outcome is negative, there is a usability problem (UP), which provokes the feeling of frustration and irritation (Bruun et al., 2016). These emotions are described as high arousal (activated) and negative valence (unpleasant) (Bruun et al., 2016), which may lead to avoiding future experiences with the product and, thus, making less informed decisions.

There has been a development of “emotional designs” that select characteristics (design shapes, colors, letters, web structure) to evoke positive or negative emotions, which can either facilitate or hinder learning in each of these platforms (Stone and University, 2005; Um et al., 2012; Heidig et al., 2015). Um et al. (2012), for example, find that positivity-inducing web designs facilitate multiple aspects of the learning process. They increase comprehension and knowledge transfer and reduce the perceived difficulty of a learning task for school students. However, as Knörzer et al. (2016) point out, there is still no conclusive evidence on how emotions influence learning in a more general population. While we advance no hypothesis in this regard, our study does include measurement of the emotional state that participants are in when they navigate through the experimental website prototypes.

We measure emotions using Noldus FaceReader™ software (den Uyl and van Kuilenburg, 2005). Individuals are videotaped during their experimental sessions with the webcam of their computers. The videotapes are later analyzed with the Facereader software. The software analyzes snapshots of the video 30 times a second and considers how closely the expressions on the face in the video correspond to profiles

Table 1
Experimental treatments. The settings with information about private pension savings and the ones with information on public benefits have the same four treatment conditions but differ in the content that is displayed under each condition.

Information Structure	Information Format		
	Text		Video
	Long-Product	Baseline	Video Treatment
	Short-Profile	Profile Treatment	Video and Profile Treatment

showing happiness, sadness, anger, fear, disgust, surprise, neutrality, and arousal, using the Facial Action Coding System (FACS). Facereader has been shown to be accurate in identifying individuals' emotions, both intended and spontaneous (Lewinski et al., 2014; Terzis et al., 2010). It has been successfully employed in a number of experimental economic studies (Fiala and Noussair, 2017; Van Leeuwen et al., 2018; Breaban and Noussair, 2018).

3. Experimental design

The experiments consist of three sections, the first is a questionnaire that sets the context of pension information and include sociodemographic questions used for block randomization. After that, participants are randomly assigned to one of the treatment websites structured as described in Table 1. In the online version of the experiment with subjects aged 50+, participants are asked which information they would like to access, retirement using (a) Private Savings or (b) Public Benefits.⁹ After they finished navigating the website they were invited to answer a comprehension questionnaire. Figure A.1 in the Appendix presents a diagram of the experimental design.

The Baseline treatment (Fig. 2) uses a text format with the product structure that was on the Pension Superintendency website at the time the experiment was designed. The Video Treatment (Fig. 3) keeps the same homepage structure but changes the most relevant elements of the content from text to video. The Profile Treatment (Fig. 4) changes the structure of the homepage from long to short and aggregates the concepts of the Baseline into four retirement profiles, but keeps all the information in text format. Finally, the Video-and-Profile Treatment (Fig. 5), incorporates both treatments with a homepage designed in terms of profiles and information provided in video format. The colors, graphical references, and number of products are the same across all treatments.

In terms of the structure, the **Product Treatments** display all the pension products of the website on the home screen, ordered in terms of relevance and equivalent across columns. A person could navigate freely across all products. They were also informed that all the information they needed to answer the questions correctly was located in the top two icons of the Product Treatments (see Section A.5 in the Appendix for online instructions). The instructions were available to participants at all times on the top left tab.

In the **Profile Treatments**, participants were presented with a set of four different types of people who would like to retire and were asked to choose one of the profiles. They were told that for all the information, they needed to answer the incentivized questions correctly was available in the profile of their choice. Participants could choose any profile, with the possibility of changing their minds and navigating across all types freely. This display strategy reduces the number of

⁹ In the laboratory version of the experiment, students were randomly allocated the description of a person that was looking to retire and asked to find an adequate retirement option for them. The description presented was consistent with the sex self-reported by the respondent, that is, if the respondent indicated they were female, the description was of a woman, and vice versa. The characteristics of the hypothetical people were consistent across sex. This was fully randomized between private and public pension websites.



Fig. 2. Homepage of the baseline treatment (Homepage of public website is shown in Figure A.2 in the Appendix).

products on the home screen, without reducing the total information that people can access. The information that participants found when they clicked on a profile was the same as what participants in the Product Treatment observed, namely the different retirement strands if they clicked on one of the four icons on the second and third rows of the home screen. Profiles were organized in terms of the four pension strands that were presented on the home screen, and the rest of the information in the Product Treatments was incorporated as links that participants could access within each profile — keeping the total set of tabs and links constant across treatments. The full content list was always available at the top right *Más información* “more information” tab.

Figs. 6 and 7 present the **Text** and **Video treatments**. The information on both was the same and simplified from that on the official Chilean Pension Authority website. They both present information in a straightforward and informative manner, explaining the main concepts (i.e. annuities) and their characteristics. The information was organized into paragraphs with a clear focus on conveying facts and details to the reader, providing explanations and examples, and concluding with an invitation for the reader to consider whether a certain concept (annuity) is suitable for their preferences. The videos were written from the perspective of a couple who wanted to retire and asked questions to an informed counselor in the form of a voice-over. For symmetry, the content in the text treatments is equivalent to the script used in



Fig. 3. Homepage of the baseline with videos treatment (Homepage of public website is shown in Figure A.3 in the Appendix).

the videos. Graphical elements of the videos, such as the male and female figures, are included as images in the text treatments, to keep the socio-identity and graphics references constant across treatments. The graphics used in the websites were designed to be inclusive, representing different geographical locations and segments of the Chilean population. The videos included sign language and subtitles for ease of access.

The content on the current institutional website is written in complex legal terms that are impractical for a tutorial aimed at simplifying pensions. To address this, the research team developed simplified content with the collaboration of experts from the Pension Superintendency and the Chilean Social Pensions Institute (IPS), ensuring alignment with legislation and an impartial presentation of the pros and cons of different alternatives. All materials were piloted with a diverse group of front-office and back-office employees, including experts from

these institutions, to ensure clarity, accessibility, and cultural relevance. Visual references to national landmarks and relatable examples further enhanced the content's appeal and usability.¹⁰

3.1. Behavioral outcomes of interest

A first behavior of interest is the likelihood that an individual completes the comprehension questions. Failure to do so would be considered as attrition to the main dependent variable of interest which is the level of comprehension. The treatments might alter the participants' willingness to navigate through the website at all, altering

¹⁰ The full content is available at the following link <https://github.com/deniselaroze/Pensions-Website-Design/tree/master/Treatments/Content>.



Fig. 4. Homepage of the profile-structured homepage with information in text treatment (homepage of public website is shown in Figure A.4 in the Appendix).

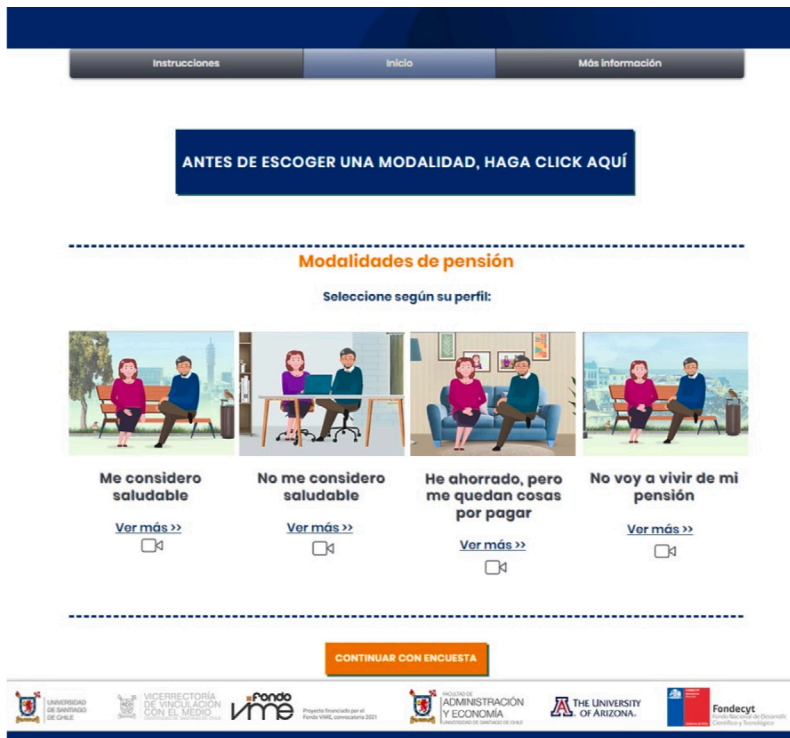


Fig. 5. Homepage profile-structured homepage with information in video treatment (homepage of public website is shown in Figure A.5 in the Appendix).

their behavior in terms of their willingness to continue with the original survey (they could just close the browser and forfeit all payment) or opt out of the comprehension questions, and get paid for finishing the survey without making any effort to learn.

The second, and main, outcome of interest is the number of correct responses to a comprehension test participants are asked to complete after navigating through the website. The comprehension tests consist

of a series of seven multiple-choice questions, each of which has one correct response. The content of the questions varies depending on the retirement system: Private Pension or Public Benefits, but is consistent in structure across treatments. Questions in the Private scheme included topics such as: (1) whether it is compulsory to retire at 65 for men and 60 for women, (2) what is the first step needed in order to start the retirement process, (3) whether everyone can choose between different

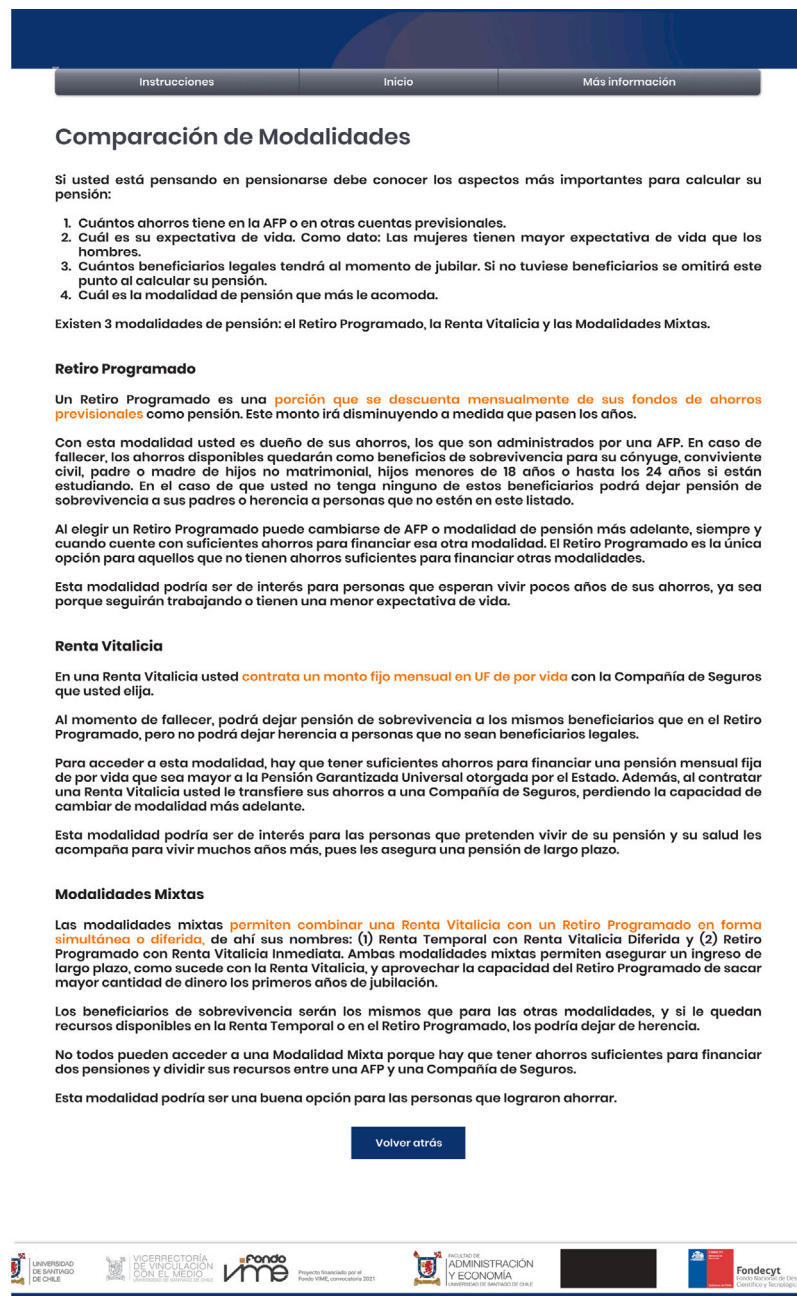


Fig. 6. Screenshot of treatments with text for the private pension scheme. The version for the public benefits website is in Appendix A.2.

retirement schemes, (4) what is the purpose of the Pension Amounts Inquiry and Offers System (SCOMP), (5) whether one can make more than one inquiry to the SCOMP, (6) if hiring a consultant is compulsory and (7) who gets ownership of the funds once the decision is made (the person or an insurance company).

Questions regarding public benefits included: (1) at what age can a person apply to the funds, (2) what is the first thing a person needs in order to apply, (3) who has access to the benefits (if foreigners are included or not), (4) how the public benefits are financed (with taxes or individuals' private savings), (5) what are the financial reasons why a person could be denied a benefit, (6) whether it is necessary to be retired to access pension benefits, and (7) what is the maximum amount one can receive as a pension benefit. The wording and correct responses to the questions are available in Section A.7 in the Appendix. Responses to individual questions are coded as dummies, with 1 for a correct response and 0 otherwise, and the sum of correct responses is used as

the dependent variable of interest. For more context, section A.1 in the Appendix presents a description of the Chilean pension system.

Answering the comprehension questions is incentivized, with each correct question adding 500 CLP (approx 0.60 USD) to a participant's earnings. For Private Pensions, a person could opt to (a) make the effort of answering the questions on their own and earning a maximum of 3500 CLP, if they answer every question correctly (7 questions * 500 CLP); or (b) ask for help (paying for advice) and earning 2700 pesos with certainty. The latter is an option that could be reasonable for people who expect to answer five or fewer questions correctly. For Public Pensions, a person who opted not to answer the comprehension questions earned 0 for the responses, mimicking the costs of not applying for a benefit. Participants could access the websites with the correct information at the time of responding to the comprehension questions if they wanted to. In all treatments, participants who completed the survey left with a baseline of 6500 CLP (approx 8 USD), on top of which they earned extra money for the comprehension questions.

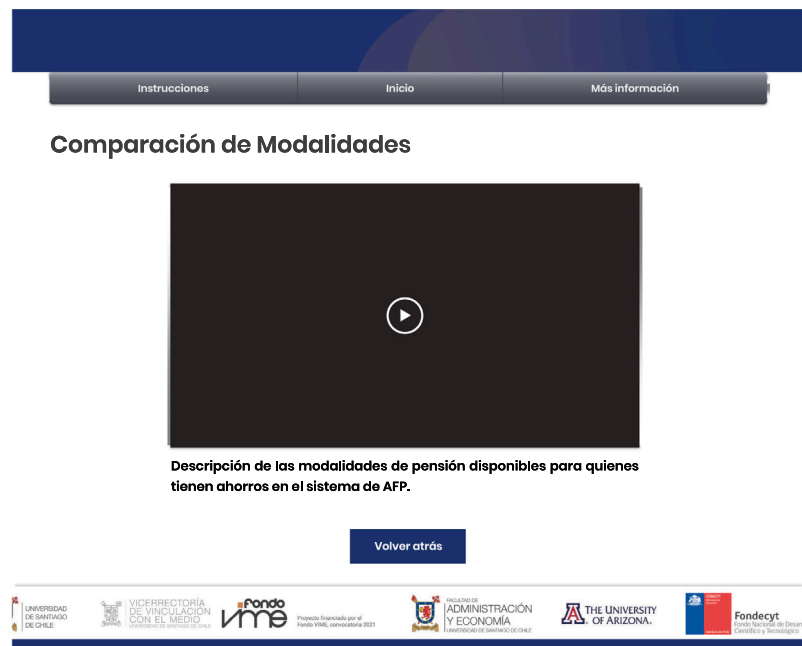


Fig. 7. Screenshot of treatments with video for the private pension scheme. The version for the public benefits websites is in Figure A.7 in the Appendix.

A third set of dependent variables of interest concerns the user's evaluation of the experience of participating in the experiment and navigating the prototypes. Participants indicated (a) if they found the information useful, (b) if they would recommend the website, and (c) the emotions they experienced during the process. These elements are relevant to understanding the overall reaction to the experience participants have during the experiment.

3.2. Robustness tests included in the design

To stress test the experimental design and ensure that results are consistent across different populations (as well as the Private and Public content variations) the experimental design includes both an online sample that represents the target population of people aged 50–70 that have not yet retired, as well as a controlled lab version with a student sample aged between 18–33 with an average age of 21.3.

The design aimed for the laboratory version to closely mirror the on-line study. Consequently, the two studies differ in only four substantive elements. (a) The lab version involved university students, facilitating replication for researchers with access to similar participants who lack strong priors about pensions. (b) Participants in the lab were randomly assigned to treatments and to a description (that matched their gender) of a person that is interested in retiring, whose interests they are asked to consider when navigating the website and gathering information (see Section A.5). (c) The laboratory setting afforded precise control over the experimental environment and participants' task engagement, ensuring alignment between intention to treat and treatment assignment. (d) In the lab sessions, consenting students were filmed and their emotions were tracked using facereading software.

To enable balance tests and to acquire demographic information, we collected data on personal characteristics such as gender, age, financial literacy, level of education, and income level (proxied by access to public or private healthcare). Previous evidence suggests that men have higher baseline levels of financial literacy on average and that there are gender differences in retirement investment decisions (Sundén and Surette, 1998; Lusardi and Mitchell, 2008, 2014). However, it is not obvious why men or women would react more to changes in a website's user interface or the inclusion of a video. Nevertheless, given the potential policy implications of this research, it is important to rule out negative effects to any subgroup of the population and test for moderating factors in any treatment effects.

3.3. Hypotheses

Given the intention with which the treatments are designed, the expectation is that they would contribute to increasing the effectiveness of the online tutorial. They would do so both by increasing the number of people that finish the tutorial, as well as the level of understanding of relevant retirement-related topics. The innovations (treatments) would also improve users' experiences.¹¹

Specifically, we hypothesize that:

- (1) More people will complete the tutorial when (a) information is provided with videos than with text and (b) information is organized into profiles.
- (2) The rate of correct responses on a comprehension questionnaire will be higher when (a) information is provided with videos than with text and (b) information is organized into profiles.
- (3) People's self-reported evaluations of the website will be higher when (a) information is provided with videos than with text and (b) information is organized into profiles.

4. Data collection

Online sample. The 55–70 year-old participants included in the online version of the experiment were recruited from the CESS Universidad de Santiago de Chile online subject pool between July 27 and August 3, 2022. Of the 904 people who consented to participate in the experiment, following pre-registered protocols, 292 were excluded because they indicated that they were already retired, and 38 participants were excluded because they were under age 50. Drop outs in the first stage of the data collection (before treatment assignment) was low, with one person leaving after the first question and another at the end of the instructions, before reaching the experimental website.

The 458 people for which there is response data are split into reasonable gender-split with 47% that self-identify as male and 53%

¹¹ At the time of pre-registration, we were worried that people would abandon the experiment halfway through. For this reason, we registered several hypotheses about attrition. Since attrition was so low, three hypotheses were aggregated into the first one, to make it easier for the reader.

Table 2Balance test: characteristics of participants assigned to each of the treatments in the online sample. *P*-value for symmetry tests in the distributions.

Variable	Baseline	Profile	Video	Profile + Video	Total	P-value
Gender						
Female	63 (25.51%)	60 (24.29%)	67 (27.13%)	57 (23.08%)	247	0.97
Male	57 (26.27%)	52 (23.96%)	55 (25.35%)	53 (24.42%)	217	0.83
Financial Literacy						
Low	36 (27.48%)	32 (24.43%)	29 (22.14%)	34 (25.95%)	131	0.85
Mid	71 (27.10%)	58 (22.14%)	71 (27.10%)	62 (23.66%)	262	0.58
High	11 (18.03%)	19 (31.15%)	19 (31.15%)	12 (19.67%)	61	0.29
Education						
Post-graduate	20 (23.53%)	18 (21.18%)	24 (28.24%)	23 (27.06%)	85	0.78
Primary/High	36 (24.83%)	32 (22.07%)	41 (28.28%)	36 (24.83%)	145	0.77
University	65 (27.31%)	63 (26.47%)	57 (23.95%)	53 (22.27%)	238	0.68
Healthcare System						
Public	76 (24.68%)	76 (24.68%)	80 (25.97%)	76 (24.68%)	308	0.98
Private	44 (28.21%)	36 (23.08%)	41 (26.28%)	35 (22.44%)	156	0.71
Age						
Mean (SD)	57.89 (4.58)	58.47 (5.49)	58.06 (4.77)	58.62 (4.50)	–	0.54

Table 3

Balance test: characteristics of participants assigned to each of the treatments in the lab sample.

Variable	Baseline	Profile	Video	Profile + Video	Total	P-value
Gender						
Female	32 (25.00%)	32 (25.00%)	33 (25.78%)	31 (24.22%)	128	0.99
Male	22 (25.58%)	22 (25.58%)	22 (25.58%)	20 (23.26%)	86	1.00
Financial Literacy						
Low	26 (25.49%)	23 (22.55%)	32 (31.37%)	21 (20.59%)	102	0.44
Mid	20 (23.81%)	24 (28.57%)	19 (22.62%)	21 (25.00%)	84	0.88
High	7 (38.89%)	4 (22.22%)	2 (11.11%)	5 (27.78%)	18	0.41
Education						
Post-graduate	1 (33.33%)	2 (66.67%)	–	–	3	0.30
Primary/High	46 (26.14%)	44 (25.00%)	43 (24.43%)	43 (24.43%)	176	0.99
University	7 (19.44%)	9 (25.00%)	12 (33.33%)	8 (22.22%)	36	0.67
Healthcare System						
Public	40 (24.54%)	41 (25.15%)	46 (28.22%)	36 (22.09%)	163	0.74
Private	14 (28.00%)	12 (24.00%)	9 (18.00%)	15 (30.00%)	50	0.64
Age						
Mean (SD)	21.15 (1.86)	22.25 (7.08)	21.68 (3.13)	22.41 (6.84)	–	0.58

as female. The sample has a high proportion of educated participants, with 30% with high school or basic education, 52% with a college or university degree, and 18% that have a postgraduate degree. As a proxy for income, 33% have a private healthcare plan.¹² The financial literacy of the sample is low, with 29% unable to answer any of the three financial literacy questions correctly (classified as Low Fin. Lit.). 58% are considered mid-level in that they answered 1 out of 3 financial literacy questions correctly (classified as Mid. Fin. Lit.), and the remaining 13% who answered 2 or 3 out of 3 questions are classified as High Fin. Lit. In this online version of the experiment, participants self-selected into the Private or Public pensions websites, with 200 (44%) self-selecting into the Public benefits, and 258 (56%) into the Private benefits.

On average participants earned 9036 CLP (9–10 USD) from their efforts in the experiment, with participants that completed the Private Pensions task earning 9149 on average and those in the Public Benefits task 8889 CLP on average, equivalent to 2.9 times the minimum hourly wage. The average completion time was 31 min, for participants that completed the survey in one continuous 120 min interval.¹³

¹² People in the top 20% of income have private health insurance, while the rest of the population mainly has access to public healthcare. A small group of military and police personnel fall into a different health insurance system.

¹³ See the descriptive statistics in the replication material.

Laboratory sample. This sample was composed of students aged between 18–33 with an average age of 21.3. The laboratory version of the study was conducted as a robustness test of our main findings. Therefore, we did not try to keep all aspects of the experiment the same as in the online sample, as we would if we were considering the subject pool as a treatment variable. 60% self-identify as female and 40% as male. The CESS Lab at Universidad de Santiago de Chile (USACH) is located within the *Faculty of Administration and Economics (FAE)* (Management and Economics). While participant recruitment primarily draws from FAE's Departments of Management, Economics, and Public Administration, the lab also attracts students from across the broader USACH community. This includes students from faculties such as Psychology and Education, providing a diverse participant pool that enhances the lab's research initiatives.

Despite being university students, financial literacy is low, with 50% unable to answer any of the three literacy questions correctly, 41% able to answer one, and 9% answering 2 or 3 correctly. Of the 222 people that started the experiment, 202 people answered the comprehension questions, 101 in the Private and 101 in the Public conditions. On average participants earned 9220 CLP from their efforts in the experiment, with participants which completed the Private Pensions task earning 9231 and those in the Public Benefits task 9208 CLP on average.¹⁴

¹⁴ The average response time was 53 min. See the descriptive statistics in the replication material.

Tables 2 and 3 present balance tables with frequencies of observable individual level characteristics. As can be observed in both the online and the lab samples, assignment to treatments is balanced by gender, financial literacy, level of education, and access to private healthcare (a proxy for income). This is in line with expectations as treatment assignment was automated through the Qualtrics functions. This result is supported by Multinomial Logistic Regression using treatment allocation as the dependent variable and characteristics of the online participants as independent variables. The results are given in Appendix Tables A.4 to A.6. Balance was also achieved within the Private Pension vs Public Benefits information subgroups, allowing for comparisons across website contents — see Tables A.4 and A.5 in the Appendix.¹⁵

5. Results

5.1. The online study with 50–70 year olds

We first consider Hypothesis (1), which asserts that the user interface would affect the likelihood of completing the tutorial. Of the 572 participants assigned to treatments, we are able to trace 458 to the third stage of the experiment where the dependent variables of interest were collected.¹⁶ The other 114 were not possible to match with the post-treatment response data. The proportion of people that left while observing the treatment (attrition) was similar across treatments, with 20% in the Baseline, 15% in the Profile Treatment, 19% in the Video Treatment, and 25% in the Video*Profile Treatment. The effect of treatment on the probability of finishing the tutorial is insignificant (see Appendix Table A.2 columns 1 and 2). Furthermore, only 11 participants opted out of the comprehension questions, 1 person in the relatively low-cost-of-opting-out Private Pension branch of the experiment, and 10 people in the high-cost-of-opting-out Public Benefits branch. Table A.2 (columns 3 and 4) also presents logit estimations of the likelihood of avoiding the comprehension questions. None of the variables included in the estimations are significant predictors of opting out of the tutorial. Thus, for the remainder of our analysis, attrition is assumed to be random. Hypothesis (1) is not supported.

The most substantive effect of the treatments is on the level of comprehension of retirement-related topics, where the addition of key information in a video format can substantively increase the number of correct responses. Fig. 8 presents the mean number of correct responses by treatment in the online sample. Average scores are high with 67% (4.7/7) correct responses in the Baseline, 65% (4.6/7) in the Profile, 80% (5.6/7) in the Video and 75% (5.3/7) in the Profile and Video combination treatment. Scores are higher in both treatments that include videos relative to the text only treatments, independently of the profile structure of the homepage. These results suggest that all treatments provide information that is comprehensible for people, but that understanding it is more effective when there is a video.

Table 4 presents OLS estimates of the number of correct responses as a function of treatment. As can be observed, the Video treatment is the main driver of increases in comprehension and its impact is significant

Table 4

The number of correct responses, as a function of treatment and literacy level (OLS estimates with heteroscedasticity robust standard errors).

	Number of correct responses					
	Full Online (1)	Public (2)	Private (3)	Female (4)	Male (5)	Does not know (6)
Profile _i	−0.180 (0.205)	−0.016 (0.306)	−0.313 (0.272)	−0.159 (0.293)	−0.224 (0.288)	−0.331 (0.364)
Video _j	0.864*** (0.165)	0.911*** (0.246)	0.854*** (0.219)	1.023*** (0.232)	0.684** (0.233)	1.108*** (0.243)
Profile _i *Video _j	−0.050 (0.262)	−0.312 (0.398)	0.153 (0.346)	−0.237 (0.363)	0.157 (0.383)	−0.022 (0.459)
Mid Fin. Lit.	0.093 (0.151)	−0.195 (0.229)	0.295 (0.199)	−0.066 (0.203)	0.271 (0.222)	−0.187 (0.255)
High Fin. Lit.	0.661** (0.203)	0.667* (0.320)	0.620* (0.258)	0.502 (0.334)	0.822** (0.264)	0.232 (0.374)
Constant	4.594*** (0.161)	4.698*** (0.251)	4.513*** (0.210)	4.605*** (0.239)	4.580*** (0.210)	4.603*** (0.264)
Observations	444	189	255	238	206	178
R ²	0.111	0.107	0.131	0.115	0.114	0.139
Adjusted R ²	0.1	0.083	0.113	0.096	0.092	0.114

Note: *p<0.05; **p<0.01; ***p<0.001.

at the $p < 0.05$ for different subgroups of the sample. The columns in Table 4 contain estimates for different subsets of the sample and address Hypothesis (2). The hypothesis stated that the rate of correct responses would be higher under the treatments with videos and profiles than in the other treatments. Column 1 includes all participants who completed the comprehension and financial literacy questions of the survey. In line with Hypothesis (2), it is possible to observe a significant effect for the Video treatment. However, the Profile treatment has no effect, which is not supportive of the hypothesis.

If the likelihood of answering all questions correctly is used as the dependent variable, following (Lusardi et al., 2017), similar patterns emerge. Table A.12 in the Appendix shows that videos are associated with a 12.3 percentage point increase in the probability of answering all questions correctly (see Table 4). This result aligns closely with the 8.3 percentage point increase in financial literacy observed in the study by Lusardi et al. (2017) from watching videos.¹⁷

The experimental design employed block randomization to ensure balanced estimations across Private and Public information websites, as well as gender comparisons. Columns 2 and 3 divide the sample into participants who viewed the Public Pension website (Column 2) and those who viewed the Private Benefit website (Column 3). In both groups, the Video treatment has a significant effect of similar magnitude. Columns 4 and 5 present results by gender, showing that the Video treatment increases the average number of correct responses for both men and women, with a larger effect observed among women. Column 6 focuses on participants who indicated uncertainty about their preferred type of pension, serving as a proxy for individuals with limited prior knowledge about pensions. This group exhibits the strongest treatment effect, with participants exposed to videos answering, on average, 1.1 more questions correctly than those exposed to text alone.

An additional subgroup analysis examines whether individuals with low or high-income levels learn more effectively when provided information tailored to their income group. Table A.14 in the Appendix indicates this is the case. Coefficients are higher for low-income individuals accessing Public Benefit information (coef. = 0.973) compared to Private Benefit information (coef. = 0.596). Conversely, higher-income

¹⁵ Power calculations, based on the observed effect sizes from the model estimations, indicate that achieving a power of 0.8 requires between 20 and 50 observations per treatment group, which is in line with this experimental dataset. This estimation assumes effect sizes ranging from 0.6 to 0.9 correct questions and a normally distributed error with a mean of 0 and a standard deviation of 1. Detailed power estimation analyses are available in the associated GitHub repository.

¹⁶ Participants could answer the comprehension questions as many times as they liked and got paid for their best attempt. 8 people answered the questionnaire more than once. To keep their observations comparable with the rest, we kept the first response of the observation in which they progressed the longest. This excludes mistakes like people who clicked on the wrong option and then went back and completed the version they had intended.

¹⁷ Three participants completed the comprehension questions but not the financial literacy questions. Including these three observations does not affect the estimations in any substantive or significant way. Table A.13 in the Appendix provides the estimations without controlling for financial literacy.

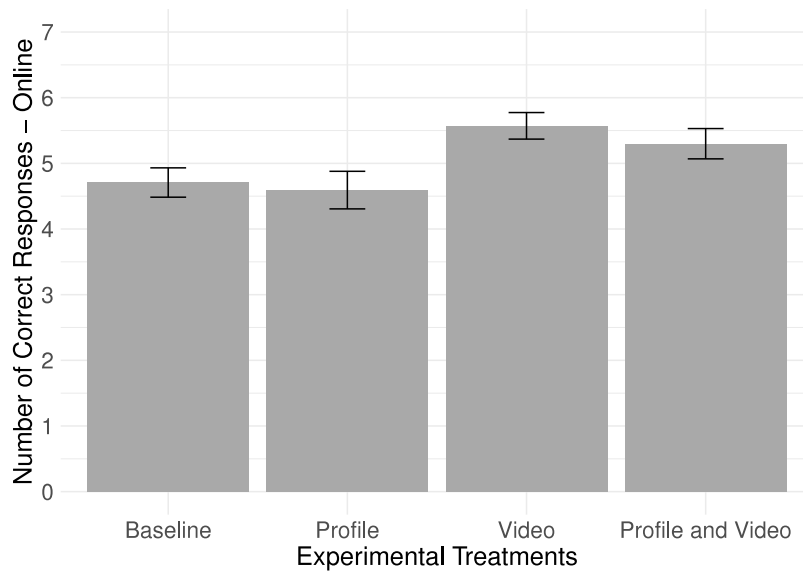


Fig. 8. Number of correct responses by treatment assignment for the online experiment, with 95% confidence intervals.

individuals learn more from the videos on the Private Pension website (coef. = 1.209) than on the Public Benefits website (coef. = 0.621).

Financial literacy was included as a control in these estimations due to its established relationship with understanding pension-related topics and standard financial literacy measures (Lusardi and Mitchell, 2011b). As expected, higher financial literacy levels are associated with higher rates of correct responses.¹⁸ The coefficient for ‘High’ financial literacy (0.661) is comparable to that of the Video treatment (0.864). However, achieving comprehension appears challenging for the 50+ population, as the benefits of financial literacy are concentrated in the ‘High’ category, which comprises only 13% of the sample. This highlights the difficulty of attaining these levels of financial literacy in this demographic.

For further analyses we used Bayesian Additive Regression Tree (BART) estimations (Dorie et al., 2022) to test for treatment heterogeneity between Private Pension vs Public Benefits information (Figures A.12 and A.13) across gender (Figures A.14 and A.15) and financial literacy (Figures A.16 and A.17). This machine learning approach for estimating conditional average treatment effects (CATEs) is non-parametric, thus avoiding the imposition of any functional form on the estimated treatment and co-variate effects. The estimates indicate that the Video treatment significantly enhances the levels of correct responses across all subgroups, with no significant interactions observed.

5.2. The lab study with students

In the lab version of the study, attrition is not a concern. Only 7 participants avoided the comprehension questions (5 in the Private and 2 in the Public condition) and thirteen observations were lost because of technical errors in the lab. As before, none of this attrition is correlated with the treatments (see Appendix Table A.3).

Fig. 9 shows the number of correct answers to the comprehension questions by treatment. It shows a similar pattern to the online study, with the Video treatment exhibiting the highest level of comprehension, with Profile+Video second.

Table 5 reports the estimates of OLS models of the determinants of the number of correct responses. They display the same pattern as in

Table 5

Number of correct responses for different sub-groups of the Lab sample (OLS estimates with heteroscedasticity robust standard errors).

	Number of correct responses				
	Full Lab (1)	Public (2)	Private (3)	Female (4)	Male (5)
Profile _i	0.187 (0.244)	0.456 (0.314)	−0.058 (0.370)	0.434 (0.307)	−0.337 (0.417)
Video _j	0.858*** (0.237)	0.838** (0.315)	0.894* (0.357)	1.105*** (0.272)	0.491 (0.435)
Profile _i *Video _j	−0.595 (0.337)	−0.900* (0.435)	−0.294 (0.509)	−0.958* (0.423)	0.018 (0.549)
Mid Fin. Lit.	0.161 (0.175)	0.087 (0.231)	0.292 (0.261)	0.058 (0.214)	0.355 (0.305)
High Fin. Lit.	1.121*** (0.241)	0.888** (0.331)	1.407*** (0.309)	1.490*** (0.329)	0.847* (0.342)
Constant	4.848*** (0.207)	4.979*** (0.279)	4.682*** (0.315)	4.704*** (0.234)	5.139*** (0.397)
Observations	200	99	101	122	78
R ²	0.112	0.111	0.135	0.146	0.097
Adjusted R ²	0.089	0.063	0.089	0.109	0.035

Note: *p<0.05; **p<0.01; ***p<0.001.

the online sample (Table 4), providing robust evidence that the Video treatment has a significant effect on learning about the retirement process. The effect is observable in the full sample, as well as the Private and Public data subsets and for women.¹⁹ The magnitude of the effect is very similar in the lab and online samples, on the order of an increase in 0.8 correct responses. As with the online sample, the male subset presents the lowest impact and is not significant in the lab subgroup. The magnitude of the coefficient (0.491) is positive and similar to the 0.684 of the online sample. Overall, these results provide robust evidence in favor of Hypothesis (2) for the video but not the profiles.

Once again, Bayesian Additive Regression Trees (BART) estimations are conducted to test for heterogeneity of the treatment effect in the student sample. The results suggest that there are no substantive

¹⁸ High financial literacy is defined as correctly answering two or more compound interest rate questions; mid-level as one correct response; and low as zero correct responses.

¹⁹ The inclusion of the Financial Literacy controls reduces the sample by 2 observations for the Private Pension estimations. Table A.12 presents the estimations without controls.

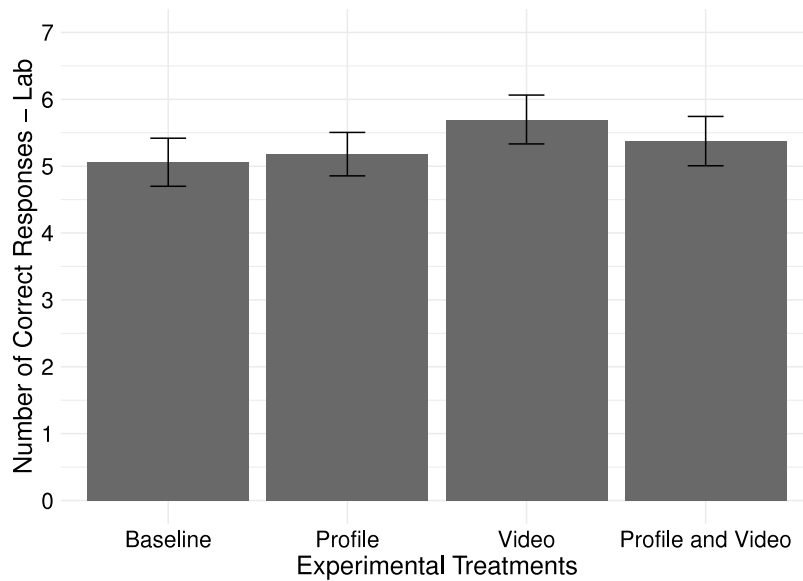


Fig. 9. Number of correct responses by treatment assignment for the laboratory data, with 95% confidence intervals.

Table 6

Comparison of OLS models on a 0–10 recommendation score, with 10 as the highest value, estimated with heteroscedasticity robust standard errors.

	NPS Lab	NPS Online
Profile _i	−0.148 (0.362)	−0.047 (0.274)
Video _j	0.212 (0.291)	0.431 (0.252)
Profile _i * Video _j	0.493 (0.468)	−0.264 (0.368)
Mid Fin. Lit.	−0.276 (0.253)	−0.451* (0.194)
High Fin. Lit.	0.306 (0.326)	−0.080 (0.262)
Constant	8.215*** (0.244)	8.347*** (0.214)
Observations	204	455
R ²	0.035	0.02
Adjusted R ²	0.01	0.009

Note: *p<0.05; **p<0.01; ***p<0.001.

Table 7

Valence and arousal experienced by participants during the website navigation as a function of treatment (OLS).

	Valence (1)	Arousal (2)
Profile _i	0.011 (0.027)	0.032* (0.018)
Video _j	0.013 (0.027)	0.029 (0.018)
Profile _i * Video _j	−0.008 (0.037)	−0.076*** (0.025)
Constant	−0.146*** (0.020)	0.255*** (0.014)
Observations	117	117
R ²	0.003	0.083
F Statistic (df = 3; 113)	0.129	3.388**

Note: *p<0.1; **p<0.05; ***p<0.01.

interactions of the treatments with gender or financial literacy, but it appears that the students that were randomly assigned to the Private Pension website gained more from the Video treatments than those that observed the simpler Public Benefit information — see Figures A.12 to A.17 in the Appendix.

5.3. Self-reported evaluations of the tutorial and emotional responses

Hypothesis (3) proposed that the Profile and Video treatments would improve people's self-reported evaluations of the tutorial. However, Table 6 suggests there is no difference in how much people would recommend a website across treatments. This was measured with responses to the question 'Considering your experience on this website, how likely are you to recommend it to a family member or friend?' answered on a scale from 0–10. As shown in tables A.7 and A.8 in the Appendix, the mean levels of recommendation are high, at approximately 8.5 out of 10 in the lab and 8.3 in the online subject pool, but there are no significant differences across treatments.

One possible reason why there are no treatment effects is that the content was consistent for all treatments (within information groups)

and was written in a citizen-friendly language that is substantively easier to understand than the status quo. The positive evaluations of the website are also present in the responses to the questions "How easy was it to find the information you were looking for?" (1–5 scale), where the average response is 4.4 out of 5, as well as on "Is the information available on the website sufficient to make a decision?" (0–10 scale), where responses averaged 7.9 out of 10.²⁰

The literature on user experience and user interfaces suggests that a positive experience can enhance understanding (Stone and University, 2005; Um et al., 2012; Heidig et al., 2015). Therefore, it is relevant to understand how changes in a website structure alter the emotional dimension of user experience. To do so, in the lab sessions, we recorded videos of the students who consented and processed them using the Noldus FaceReader emotional analysis software.

Table 7 presents Ordinary Least Squares (OLS) estimates of the levels of valence and arousal participants experienced while they navigated. In Column 1 it can be observed that the level of positive versus negative emotions (valence) is not altered by the treatments. On average negative emotions prevail in the sample (negative constant) and neither the change in the user interface nor the inclusion of videos can explain any difference across participants. However, arousal (column

²⁰ Tables A.7 and A.8 in the Appendix present the OLS estimations.

2) is negatively associated with the Video+Profile treatment combination. A possible interpretation for this reduction in arousal is that participants reduced their level of concentration under Profile+Video.

6. Discussion

This study examines the potential for enhancing pension literacy through two treatments implemented on an institutional website. The first treatment involves modifying the informational format, transitioning from text-only to a combination of text and key information presented in video form. The second treatment focuses on altering the homepage structure from a lengthy product-oriented approach to a more concise profile-based layout. These treatments were evaluated using two distinct content variations. One set of experimental treatments provided guidance on retirement using private savings, while the other set offered information about accessing public benefits. Additionally, for robustness and replicability of results (cf. [Camerer et al., 2019](#); [Duch et al., 2020](#), among others) the study engaged two different subject pools: an online subject pool representing the policy-relevant population, and a separate subject pool comprising university students that permits easy replication. Remarkably, the results remain highly consistent with both samples.

As predicted, the videos substantively and significantly improved the comprehension of both the private and public retirement processes, increasing the average number of correct responses by 0.8 out of 7. This effect is consistent across the different topics on retirement that we queried. The effect sizes of changing the format from text to video are comparable to the effects of video format observed by [Lusardi et al. \(2017\)](#) or of moving from being able to answer 0 of 3 to 2 of 3 questions about compound interest correctly. In the Private Pensions retirement context, the amount of information is substantively greater than that of the Public Benefits, with videos that lasted 6:42 (Private Retirement options) and 5:00 min (how to retire with private funds) versus 4:14 (Public Benefits available) and 3:54 (how to apply to a benefit). This highlights the value of the ([Johnson et al., 2012](#)) recommendations about structuring and describing options in a simple, organized way, but also addresses the issue of focusing on the experience of the users by making a post-evaluation decision about the optimal architecture. The results are also consistent across subject pools, with people aged 50+ doing an experiment online presenting the same effects as students in their late teens and early 20s with very little interest in pensions in the controlled environment of the laboratory.

Contrary to our expectations, changing the structure of the homepage did not contribute to any of the behaviors of interest. It did not get more people to complete the tutorial, nor did it help comprehension, despite organizing information in a hierarchical structure and reducing the amount of unnecessary information — understood as information that is not necessary for completing the comprehension test. Furthermore, the profile structure, in combination with videos, reduced the level of arousal experienced by students in the lab. While it is tempting to interpret the decrease in arousal as reflecting lower concentration that leads to worse performance, the evidence on the relationship between arousal and performance is mixed. The Yerkes–Dodson Law ([Yerkes and Dodson, 1908](#)) asserts that performance and arousal are positively correlated until the arousal level reaches a certain point, beyond which the correlation turns negative. Empirical evidence has tended to support this inverted U-shaped relationship ([Robazza et al., 1998](#); [Anderson, 1994](#), [Watters et al., 1997](#)).

Our efforts to simplify the presentation with profiles was not successful in improving comprehension. We view this as an illustration of the principle that simplicity of presentation does not necessarily translate to better understanding. While simplification surely improves understanding in many cases, it may not be a general rule.

One reason that the profiles may not have been effective might be that people did not relate to the profiles that were constructed. As [Johnson et al. \(2012\)](#) and [Krijnen et al. \(2017\)](#) suggest, the reduction

of the choice set into profiles has to make sense to the people that use them, and might produce misunderstandings if they do not. The research team approached the profile construction with a lot of care and knowledge about the different alternatives. However, it is indeed a difficult task to specify the appropriate profiles, especially in the Private retirement setting. Nevertheless, in the Public Benefits websites, the categories are clear-cut and non-overlapping as the profiles were organized in terms of non-compatible benefits, and the profile-structure treatments still did not affect any of the behaviors of interest.

This study shows that it is possible to help people understand complex retirement-related topics by incorporating videos that explain key elements of the process, namely the alternatives available and how to access them. This is a simple and relatively cheap alternative to increase comprehension and facilitate good decisions. The interventions are scalable and thus can potentially affect a large number of people. Even a small improvement in comprehension, if it translates to better pension decisions, given the stakes in each decision and the large number of individuals affected, mean that the interventions can yield substantial welfare improvements.²¹ Indeed, the videos produced for this experiment were made public on the Chilean Pensions Superintendency website in 2023.

CRedit authorship contribution statement

Denise Laroze: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Charles N. Nossair:** Writing – review & editing, Writing – original draft, Conceptualization. **Gabriela Fajardo:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Ximena Quintanilla:** Project administration, Funding acquisition, Conceptualization. **Paulina Granados-Zambrano:** Project administration, Funding acquisition, Conceptualization. **Pedro Vallette:** Project administration, Funding acquisition, Conceptualization. **Mauricio López-Tapia:** Software, Data curation.

Declaration of competing interest

All authors declare that they have no financial interest in the outcome of this research. All procedures in the study received approval from the relevant institutional review boards.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jbef.2025.101046>.

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²¹ The cost of producing the videos was approximately 6000 USD, while conducting the experiment, including incentive payments, amounted to 7000 USD. The videos have already been viewed more than 6000 times, translating to an initial cost of 1 USD per view across the four videos. Given that more than 160,000 individuals reach retirement age annually in Chile, the potential reach of these videos is extensive. As the videos are a one-time production cost, their cost per view will continue to decrease as the reach expands, making them an economically scalable solution.

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